

Grammar Competition and Contact-Induced Instability in the Old-to-Middle English Transition: A Computational Case Study Using the Symmetric Language Dynamical Equation

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Abstract

The transition from Old English to Middle English unfolded amid regional dialect competition, internal change, and sustained Scandinavian contact in parts of England. This paper asks whether such contact can be represented, cautiously and qualitatively, as a temporary destabilizing perturbation in grammar competition. We implement the fully symmetric language dynamical equation analyzed by Mitchener (2003), in which population frequencies of idealized grammars evolve under communicative selection and imperfect learning. The case study uses four Old English dialectal components—West Saxon, Mercian, Northumbrian, and Kentish—and introduces an idealized Old Norse component during a contact phase. Five scenarios compare a no-contact baseline with conservative, moderate, strong-contact, and canonical Mitchener parameter regimes, varying learning fidelity q , average intelligibility α , and the Old Norse contact fraction. Numerical integration with adaptive Runge–Kutta methods is evaluated through frequency trajectories, normalized entropy, the concentration statistic M_2 , and transient displacement integrals relative to the no-contact baseline. Across the historically motivated parameter portfolio, contact produces a measurable transient displacement toward a more distributed state, but not a different long-run attractor. The study is therefore exploratory and methodological: it does not reconstruct demographic history or claim that Old Norse contact caused Middle English simplification, but shows how the symmetric language dynamical equation can formalize and test qualitative hypotheses about contact-induced instability.

Keywords: language dynamics; historical linguistics; Old English; Middle English; Old Norse; bifurcation; replicator–mutator dynamics; computational modeling; language contact

1 Introduction

The transition from Old English to Middle English is one of the central episodes in the history of English. It involved large-scale changes in morphology, syntax, lexical composition, and regional variation. Among the historical forces usually discussed are internal developments, dialectal variation, Scandinavian contact in the Danelaw, and later Norman French influence. The present paper focuses only on a deliberately narrow question: how can an existing dynamical-systems model of language competition be used to explore the qualitative effect of a contact-induced reduction in learning reliability?

The mathematical starting point is the language dynamical equation developed by Komarova et al. (2001) and analyzed in detail by Mitchener (2003). Mitchener studies the fully symmetric

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case, where all grammars are interchangeable in the sense that cross-grammar intelligibility is represented by a single parameter. In that setting, he rigorously locates fixed points, analyzes their stability, and describes bifurcation scenarios as the reliability of language learning changes.

Crucially, Mitchener (2003, p. 284) explicitly suggests the transition from Old English to Middle English as a motivating historical case: Scandinavian speech may have introduced enough linguistic noise to reduce learning reliability, thereby destabilizing a previously coherent equilibrium. This paper does not claim to introduce a new mathematical model. Instead, it operationalizes that suggestion as a reproducible computational case study. The contribution is therefore methodological and exploratory: we define historically interpretable scenarios, simulate the symmetric language dynamical equation across contact phases, and evaluate how sensitive the resulting dynamics are to assumptions about learning fidelity, average intelligibility, and the idealized strength of the Old Norse contact component.

The scope of the claim is intentionally limited. The model tracks frequencies of idealized grammatical systems, not specific morphosyntactic features. It cannot, by itself, prove that Old Norse caused the morphological simplification of Middle English. Rather, it provides a controlled mathematical environment in which one can ask how contact-driven reductions in learning fidelity affect the stability of coherent and incoherent population states.

2 Related Work

Mathematical approaches to language change have long used tools from evolutionary dynamics, learning theory, and population-level modeling. Niyogi and Berwick (1997) proposed a dynamical-systems account of language change in a population of learners. Komarova et al. (2001) introduced an evolutionary model of grammar acquisition combining communicative fitness and imperfect learning. Mitchener’s later analysis provides a complete bifurcation study of the fully symmetric version of that model (Mitchener, 2003).

The present work is closest to Mitchener’s fully symmetric language dynamical equation. The model combines a replicator-like selection term with mutation-like learning error, resembling quasispecies dynamics (Eigen, 1971). In the symmetric case, learning reliability q becomes the main bifurcation parameter: low values of q favor incoherent equilibria, while high values can support coherent equilibria dominated by one grammar.

A related but distinct line of work applies learning-theoretic models to historical English. Niyogi and Berwick (2009) discuss language acquisition and change in a population setting and explicitly model aspects of the evolution of Middle English. Their framework differs from the present one: the focus here is not to replicate the same empirical model, but to explore the behavior of Mitchener’s symmetric replicator–mutator system under Old-English-inspired historical scenarios.

More recent work on language contact and change includes models of contact-induced bifurcation and sociolinguistic population dynamics (Kauhanen, 2020, 2022). These studies motivate the idea that contact conditions can alter the qualitative stability of linguistic systems, although they use different modeling assumptions. Earlier extensions of the language-dynamical tradition also consider competitive exclusion, coexistence, and game dynamics with learning (Mitchener and Nowak, 2003; Mitchener, 2007). Historical and corpus-oriented work on English–Norse contact provides a complementary empirical perspective on koineization, simplification, and contact-induced change (Dawson, 2003; Warner, 2017; Walkden et al., 2023). The present study should therefore be read as a computational bridge between the symmetric language dynamical equation and a historically motivated case study, not as a replacement for corpus-based historical linguistics.

3 The Symmetric Language Dynamical Equation

Consider a population in which each individual uses one of n candidate grammatical systems $G_1, \dots, G_n$. Let $x_j(t)$ denote the proportion of the population using grammar G_j at time t , with

$$x_j(t) \geq 0, \quad \sum_{j=1}^n x_j(t) = 1.$$

Thus the state vector $x(t)$ lies in the simplex S_n .

Following Mitchener (2003), the communicative fitness of grammar G_i is defined by

$$F_i(x) = \sum_{k=1}^n \frac{A_{ik} + A_{ki}}{2} x_k, \quad (1)$$

where A_{ik} represents the probability that a sentence produced by a speaker of G_i can be parsed by a speaker of G_k . The average fitness is

$$\phi(x) = \sum_{k=1}^n F_k(x) x_k. \quad (2)$$

Learning is represented by a row-stochastic matrix Q , where Q_{ij} is the probability that a teacher using G_i produces a learner using G_j . The general language dynamical equation is

$$\dot{x}_j = \sum_{i=1}^n F_i(x) x_i Q_{ij} - \phi(x) x_j. \quad (3)$$

Because Q is row-stochastic, the total population mass is conserved:

$$\frac{d}{dt} \sum_j x_j = 0.$$

Moreover, if $x_j = 0$, the right-hand side is non-negative, so the positive simplex is invariant.

3.1 The fully symmetric case

The fully symmetric model assumes

$$A = \begin{pmatrix} 1 & a & \cdots & a \\ a & 1 & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} q & u & \cdots & u \\ u & q & \cdots & u \\ \vdots & \vdots & \ddots & \vdots \\ u & u & \cdots & q \end{pmatrix}, \quad u = \frac{1-q}{n-1}. \quad (4)$$

Here a represents average intelligibility between different grammars, while q represents the probability of faithful learning. Under these assumptions, Mitchener (2003) obtains the reduced equation

$$\dot{x}_j = (1-a) \left[(q-u)x_j^2 + uM_2 - x_jM_2 \right] - aunx_j + au, \quad (5)$$

where

$$M_2 = \sum_{i=1}^n x_i^2. \quad (6)$$

Mitchener's analysis shows that the model can converge either to incoherent equilibria, where several grammars coexist, or coherent equilibria, where one grammar dominates. The present study uses this reduced equation as the computational object of study.

4 Historical Parameterization

The historical parameterization used in this study is exploratory. It is not a demographic reconstruction of early medieval England. The variables x_j are interpreted as idealized population-level weights for grammatical or dialectal systems, not directly observed proportions of speakers. Likewise, the parameters q and a are not measured historical quantities. They are scenario parameters introduced to probe the qualitative behavior of the model.

We consider four Old English dialectal components:

- G_1 : West Saxon,
- G_2 : Mercian,
- G_3 : Northumbrian,
- G_4 : Kentish.

During the contact phase, a fifth component is introduced:

- G_5 : Old Norse / Anglo-Scandinavian contact component.

The following evidence supports the parameterization, though it must remain explicitly provisional. Several useful claims are supported only indirectly in the current evidence table, for example through secondary summaries or through sources cited by other works. For this reason, the values below are implemented as sensitivity scenarios rather than as historical estimates. The strongest evidence concerns written standardization and literary prestige; the weakest evidence concerns numerical values for learning fidelity and mutual intelligibility.

Late West Saxon is treated as a proxy for written prestige and manuscript standardization, especially in connection with the late Old English / Winchester written tradition (Gneuss, 1972; Hofstetter, 1988; Gretsche, 2001). This modelling choice does not imply that West Saxon was demographically dominant. It only reflects the disproportionate role of West Saxon in the surviving written record. Mercian is included with comparable weight because the Anglian Midlands are commonly treated as central to later English developments, even though the Old English textual record does not preserve Mercian material in direct proportion to its likely territorial and sociolinguistic relevance (Hogg, 1992; Baugh and Cable, 2012). Northumbrian is assigned an intermediate initial weight because of its early cultural importance and later documentary disruption. Kentish is assigned a lower weight because of its more geographically restricted role, although this should not be interpreted as evidence that Kentish is linguistically irrelevant.

The Old Norse component should also be interpreted cautiously. The Danelaw and related Anglo-Scandinavian settlement provide historical motivation for a contact phase, but they do not yield a direct numerical estimate of the proportion of Old Norse speakers. Archaeological and historical work has proposed estimates on the order of 20,000–35,000 Scandinavian settlers in some reconstructions (Kershaw and Røyrvik, 2016), while more recent work emphasizes that the Danelaw should not be treated as a simple, unified demographic block (Raffield, 2020). In the model, the Old Norse fraction is therefore a contact-intensity parameter rather than a literal population share.

The parameter q represents learning fidelity. During the contact phase, lower values of q are used to model a noisier learning environment, where children may receive less consistent cues from a multilingual or multidialectal input distribution. This interpretation is consistent with the mechanism proposed by Mitchener (2003), but the numerical values of q are heuristic. Contact is treated here as a potential accelerator of ongoing internal changes rather than as a single external cause (Dawson, 2003; Warner, 2017), and the simulations therefore test contact as a destabilizing factor, not as a proof of causation.

Likewise, a is treated as an average cross-grammar intelligibility parameter. Work on Old Norse–Old English contact is relevant because it discusses linguistic relations and possible mutual intelligibility between speakers of the two related Germanic languages (Townend, 2002). However, this does not justify a single measured value of a . Because the fully symmetric model cannot represent pairwise differences in intelligibility among West Saxon, Mercian, Northumbrian, Kentish, and Old Norse, a is varied across scenarios rather than fixed as a historical fact.

4.1 Evidence table and modelling implications

Table 1 summarizes the current source-based interpretation. The entries should be read as modelling justifications, not measurements.

Table 1: Historical evidence, modelling implications, and evidence quality.

Component / parameter	Source-grounded evidence	Modelling implication	Evidence quality
West Saxon	Late West Saxon / Winchester written standard; strong role in manuscript transmission (Gneuss, 1972; Hofstetter, 1988; Gretsche, 2001)	High initial literary-prestige weight; not demographic majority	Strong support
Mercian	Midlands / Anglian relevance to later English developments (Hogg, 1992; Baugh and Cable, 2012)	Comparable initial weight to West Saxon in central scenarios	Moderate support
Northumbrian	Early cultural importance and later documentary disruption in northern records (Hogg, 1992; Baugh and Cable, 2012)	Intermediate initial weight; corpus-loss caveat	Moderate support
Kentish	More geographically restricted dialectal component (Hogg, 1992; Baugh and Cable, 2012)	Lower initial weight in most scenarios, not zero relevance	Moderate support
Old Norse / Anglo-Scandinavian	Danelaw settlement and Anglo-Scandinavian contact; demographic estimates remain contested (Kershaw and Røyrvik, 2016; Raffield, 2020)	Contact-intensity component, not literal speaker proportion	Moderate support
q during contact	Mixed input/contact may reduce cue consistency; modeled after Mitchener’s reliability parameter (Mitchener, 2003)	Reduced in contact scenarios; must be sensitivity-tested	Heuristic/sensitivity
a	Possible partial OE–ON intelligibility and relatedness, but not identity (Townend, 2002)	Average intelligibility varied across scenarios	Heuristic/sensitivity

4.2 Scenario design

The simulations use five exploratory scenarios, summarized in Table 2. The values should be treated as historically motivated hypotheses to be refined through source-based work and corpus

proxies. The central scenario gives equal initial weight to West Saxon and Mercian, lower weight to Northumbrian, and the smallest weight to Kentish. This choice reflects a cautious compromise between textual prestige, territorial relevance, and the absence of direct demographic data.

Table 2: Exploratory parameter scenarios. Values are not historical measurements; they define sensitivity regimes. The canonical comparison is a methodological reference point based on Mitchener’s commonly illustrated $a = 0.5$ regime, not a historically calibrated scenario. In the no-contact scenario, the contact phase retains $q_2 = q_1 = 0.90$ (no reduction in learning fidelity) with $n = 4$ grammars throughout; the “–” in column q_2 indicates the absence of a contact-phase reduction, not the absence of a phase.

Scenario	WS	Merc.	North.	Kent.	q_1	q_2	q_3	a	ON frac.
Conservative	0.35	0.35	0.20	0.10	0.90	0.75	0.85	0.80	0.15
Moderate	0.35	0.35	0.20	0.10	0.90	0.65	0.80	0.80	0.25
Strong contact	0.30	0.35	0.20	0.15	0.90	0.55	0.75	0.75	0.33
Canonical comparison	0.35	0.35	0.20	0.10	0.90	0.60	0.80	0.50	0.33
No contact	0.35	0.35	0.20	0.10	0.90	–	0.90	0.80	0.00

The contact scenarios are organized into three phases:

1. **Pre-contact phase:** four Old English dialectal components, high learning fidelity q_1 .
2. **Contact phase:** Old Norse is introduced as an idealized fifth component, and learning fidelity is reduced to q_2 .
3. **Post-contact phase:** the Old Norse component is removed and redistributed among Old English components, while learning fidelity partially recovers to q_3 .

The no-contact scenario is a counterfactual baseline in which no Old Norse component is introduced and learning fidelity remains at q_1 throughout all three phases. To ensure this is a strict four-grammar baseline, the integration uses $n = 4$ in all phases; the G_5 column is absent rather than initialized at zero, preventing the mutation term from generating a phantom Old Norse component.

4.3 Interpretive scope

Several aspects of the model design carry important interpretive limits. The initial West Saxon weight reflects written-prestige assignment in the surviving manuscript record, not a claim about spoken demographic dominance. The Old Norse fraction is a contact-intensity parameter calibrated to historical plausibility rather than a literal population share; similarly, the learning-fidelity values are scenario-dependent heuristics, not empirically measured quantities. The simulations treat contact as one potential destabilizing factor alongside internal change, regional variation, and later external influences, and the results bear on the transient dynamics of grammatical competition rather than on the ultimate cause of Middle English morphological simplification. Because the fully symmetric model assigns uniform pairwise intelligibility, the parameter a should be understood as a scenario sensitivity variable rather than a measured linguistic fact about any specific language pair.

5 Numerical Method

The initial value problem is

$$\frac{dx}{dt} = f(x; q, a, n), \quad x(t_0) = x_0, \quad (7)$$

where f is defined componentwise by Equation (5). Numerical integration was performed using `scipy.integrate.solve_ivp` with the explicit adaptive Runge–Kutta method RK45, tolerances `rtol`= 10^{-10} and `atol`= 10^{-12} . For each run, the following diagnostics were monitored:

- conservation of mass: $\max_t |\sum_j x_j(t) - 1|$;
- positivity: $\min_{j,t} x_j(t)$;
- normalized entropy:

$$H_{\text{norm}}(x) = -\frac{1}{\log n} \sum_{j=1}^n x_j \log x_j; \quad (8)$$

- concentration/coherence proxy:

$$M_2(x) = \sum_{j=1}^n x_j^2. \quad (9)$$

Entropy is high when several grammars coexist with comparable weights. The statistic M_2 is low near the uniform state and increases as the population concentrates around fewer grammatical systems. These quantities do not measure historical linguistic complexity; they measure diversity and concentration in the model state. Because n changes from 4 to 5 at the onset of the contact phase and back to 4 afterwards, the entropy and M_2 plots show structural discontinuities at both phase boundaries that reflect the change in model dimension, not changes in contact dynamics. Entropy is normalized by $\log n$ and the minimum possible M_2 shifts from $1/4$ to $1/5$ during the five-component contact phase.

Transient displacement integrals. To quantify the cumulative deviation of each contact scenario from the no-contact baseline, two scalar diagnostics are defined. Let $[t_0, t_f] = [0, 120]$ denote the full simulation window in model units.¹ For each contact scenario, the absolute difference between its diagnostic trajectory and the corresponding no-contact baseline trajectory is integrated over this window:

$$D_H = \int_{t_0}^{t_f} |H_{\text{norm,contact}}(t) - H_{\text{norm,no-contact}}(t)| dt, \quad (10)$$

$$D_{M_2} = \int_{t_0}^{t_f} |M_{2,\text{contact}}(t) - M_{2,\text{no-contact}}(t)| dt. \quad (11)$$

These integrals summarize how far, and for how long, a contact scenario departs from the no-contact baseline in a single scalar value: a scenario that produces a large instantaneous deviation, or one that produces a smaller but long-lasting deviation, will yield a larger integral. Because all historically motivated scenarios converge to the same final state as the no-contact baseline, D_H and D_{M_2} capture the transient geometry of the contact effect rather than any change in the long-run attractor. Numerical evaluation uses the trapezoid rule applied to the concatenated time grid, which is identical across all scenarios by construction.

A fixed-step classical RK4 implementation was used to verify convergence: comparing step sizes h , $h/2$, and $h/4$ for Phase 1 of the moderate scenario yields a maximum L_∞ error of 1.6×10^{-12} at $h = 0.1$, confirming fourth-order convergence of the reference RK4 implementation and providing indirect corroboration of the RK45 solution.

6 Computational Experiments

The experiments are organized around three questions:

¹The time variable t in the symmetric language dynamical equation is dimensionless. No specific mapping to calendar years or linguistic generations is assumed; the model explores qualitative dynamics rather than quantitative historical reconstruction.

1. How do grammar-frequency trajectories differ between contact and no-contact scenarios during and after the contact phase?
2. How do entropy and M_2 respond to the reduction in learning fidelity during contact, and how long does that response persist into the post-contact phase?
3. Across what range of (a, q_2) does contact-phase disruption remain qualitatively similar to the historically motivated scenarios, and where does the model enter a regime with stronger final-state consequences?

6.1 Temporal trajectories

Figure 1 shows the trajectory of the moderate scenario. The vertical dashed lines mark the beginning and end of the contact phase. During contact, the Old Norse component is introduced and the learning reliability parameter is reduced. The interpretation should be qualitative: the figure illustrates how the dynamical system responds to a parameter shock, not how actual speaker populations evolved year by year.

During the contact phase, the four Old English components are displaced from their pre-contact trajectories as the introduced Old Norse component and the reduced learning fidelity jointly increase diversity and reduce concentration. After the contact phase ends and q partially recovers, the Old English components consolidate again. The transient displacement—the departure from the no-contact trajectory and the path taken through state space during the contact phase—is the primary object of interest in the results below, since all scenarios converge to the same near-uniform final state.

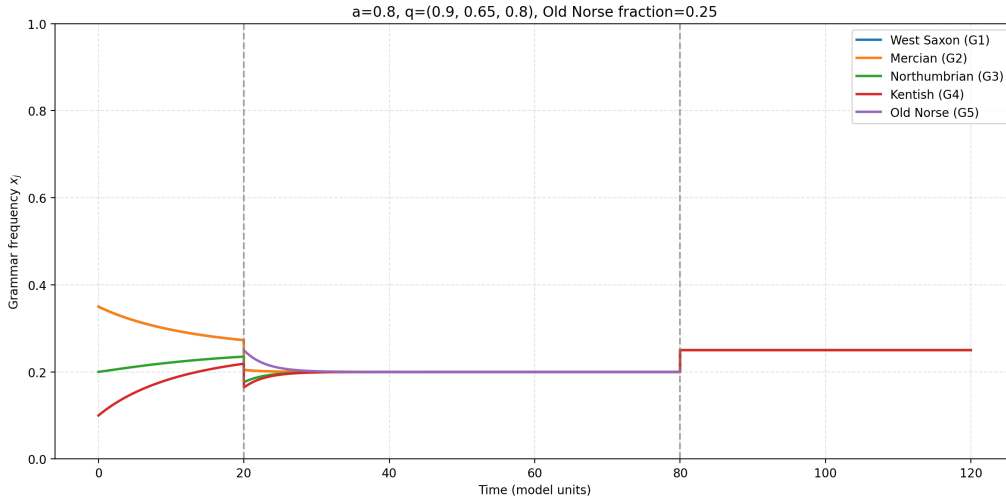


Figure 1: Frequency trajectories in the moderate contact scenario. The Old Norse component is introduced during the contact phase and removed in the post-contact phase.

6.2 Entropy and concentration

Figures 2 and 3 show model-state diagnostics for the moderate scenario. In the moderate scenario, normalized entropy rises and M_2 falls at the onset of the contact phase, indicating that the system moves toward a more distributed state. This is the expected response to a reduction in learning fidelity: when q decreases, the mutation-like contribution becomes relatively stronger and the system is pulled toward the near-uniform equilibrium. When q partially recovers in the post-contact phase, the system begins to consolidate again. The key observation is that this disruption is a phase-dependent transient: in the parameter regime used here, the final state of the system is the same near-uniform configuration regardless of whether contact occurred,

because the simulated trajectories in all historically motivated scenarios converge to the near-uniform state. The contact phase produces a detour through a more distributed region of state space, not a shift in the long-run attractor.

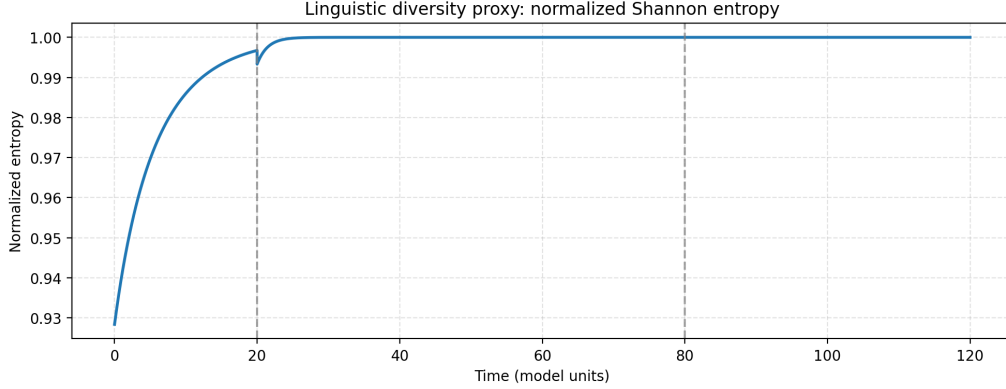


Figure 2: Normalized entropy in the moderate contact scenario. Higher values indicate greater diversity among model components. The apparent phase-boundary shifts also reflect the change in model dimension from $n = 4$ to $n = 5$ and back to $n = 4$.

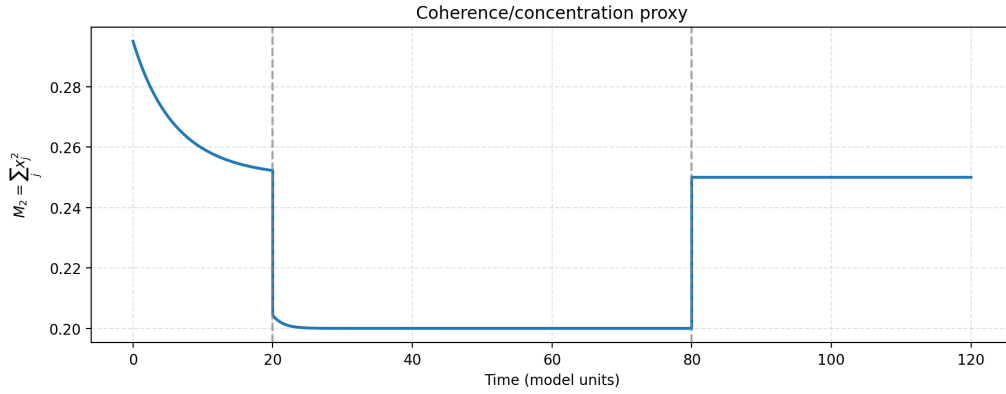


Figure 3: Concentration statistic $M_2 = \sum_j x_j^2$ in the moderate contact scenario. Higher values indicate stronger concentration in fewer components. The minimum possible value changes from $1/4$ to $1/5$ during the five-component contact phase.

6.3 Scenario comparison

Scenario comparisons allow us to avoid over-interpreting a single parameter set. The robust result across all contact scenarios is an elevation of normalized entropy and a corresponding depression of M_2 during the contact phase, relative to the no-contact baseline. The depth and duration of this transient response scale with contact intensity: greater Old Norse fraction and larger reductions in q_2 produce stronger and more persistent departures from the no-contact trajectory. Figures 4 and 5 show this pattern.

Table 3 reports final diagnostic values for the scenario portfolio. All scenarios converge to normalized entropy of 1.0 and $M_2 = 0.25$, corresponding to the uniform four-component distribution. Final-state divergence is therefore not a feature of this parameter portfolio; the main differences among scenarios are transient and phase-dependent.

Figure 6 provides a direct visual comparison of the no-contact baseline and the moderate contact scenario at the level of individual grammar frequencies. In the no-contact panel, the four Old English dialects drift slowly and monotonically toward the uniform state under constant high

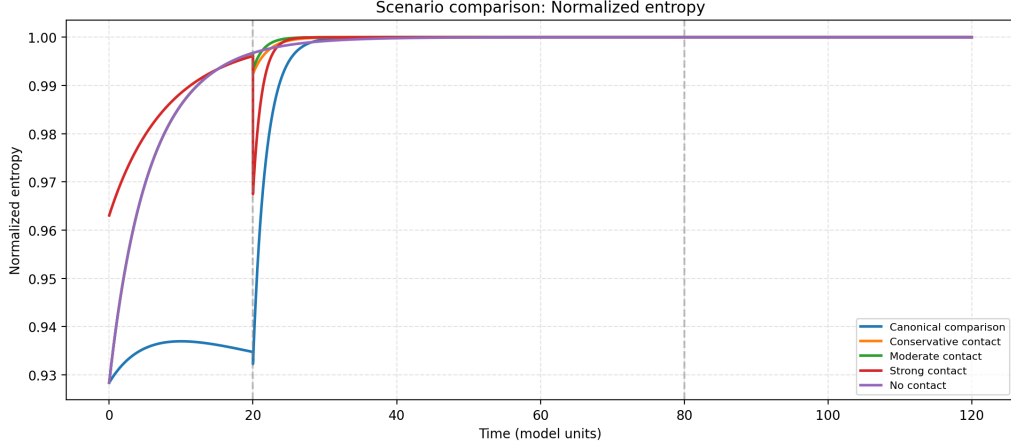


Figure 4: Comparison of normalized entropy across exploratory scenarios. Contact scenarios contain a five-component middle phase, whereas the no-contact baseline remains four-dimensional throughout.

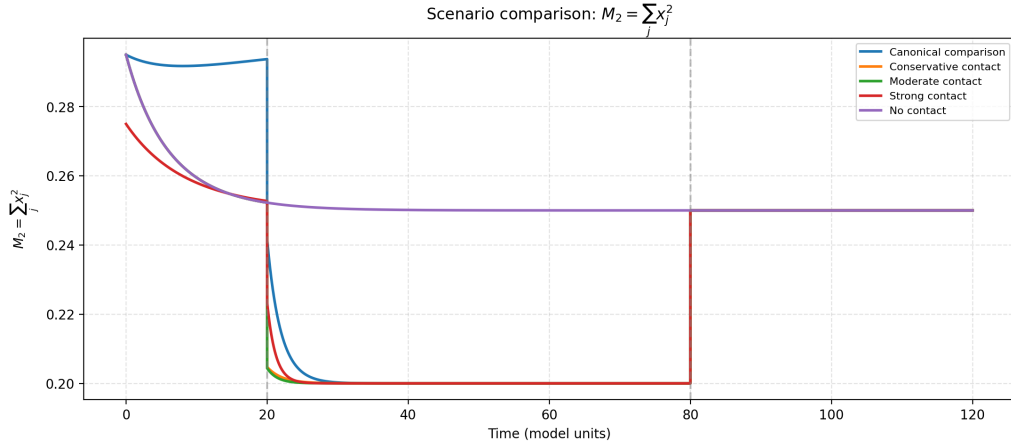


Figure 5: Comparison of M_2 across exploratory scenarios. Contact scenarios contain a five-component middle phase, whereas the no-contact baseline remains four-dimensional throughout.

learning fidelity. In the contact panel, the same four dialects are visibly displaced during the contact phase before recovering. This comparison makes the transient character of contact-induced disruption concrete: the trajectories diverge during the contact window and converge again afterward, ending at the same final state.

Table 4 quantifies this transient departure through the scalar integrals D_H and D_{M_2} defined in Equations (10)–(11). Consistent with the visual pattern in Figures 4 and 5, the integrated displacement scales with contact intensity: stronger contact scenarios accumulate a larger total deviation from the no-contact baseline over the full simulation window. The canonical comparison produces the largest displacement of the four contact scenarios because $a = 0.50$ places the system in a parameter regime where reductions in q have more pronounced dynamical consequences than in the historically motivated scenarios at $a \in [0.75, 0.80]$.

²All simulation code is publicly available at https://github.com/cmorregof/language_dynamics. The repository includes the main Python script, dependencies, and generated figures.

Table 3: Diagnostics for the exploratory scenarios, generated by the accompanying Python script. Final entropy and final M_2 are endpoint values at $t = 120$. Max mass error and minimum component are trajectory-wide extrema across all simulation phases, not final-state values.

Scenario	Final entropy	Final M_2	Max mass error	Minimum component
Canonical comparison	1.0000	0.2500	4.44×10^{-16}	0.0909
Conservative contact	1.0000	0.2500	2.22×10^{-16}	0.1000
Moderate contact	1.0000	0.2500	2.22×10^{-16}	0.1000
Strong contact	1.0000	0.2500	2.22×10^{-16}	0.1470
No contact	1.0000	0.2500	2.22×10^{-16}	0.1000

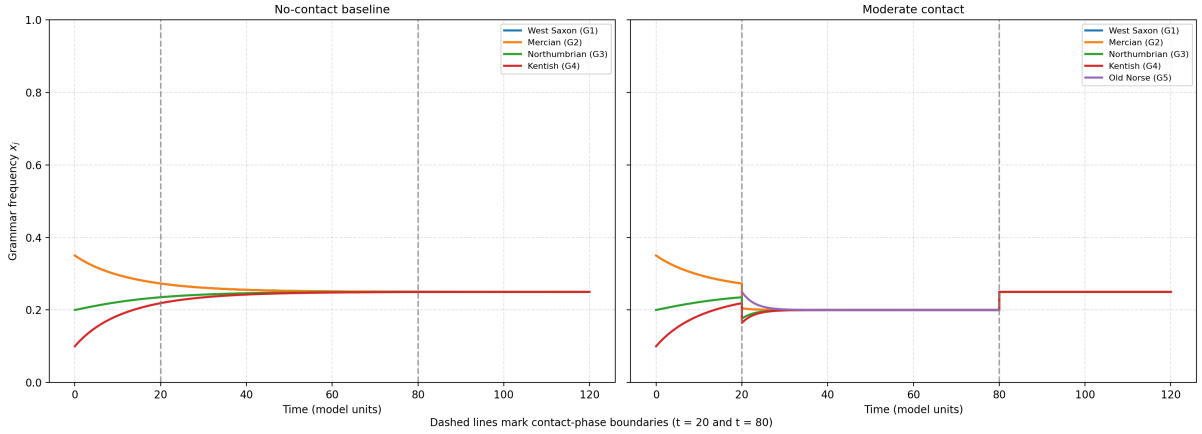


Figure 6: Grammar frequency trajectories under no-contact (left) and moderate contact (right) scenarios. In the no-contact scenario, the four Old English components evolve under constant high learning fidelity with no additional grammar introduced ($n = 4$ throughout). In the moderate contact scenario, the Old Norse component is introduced at the start of the contact phase (first dashed line) and removed at the end (second dashed line). The comparison illustrates the transient character of contact-induced disruption: trajectories diverge from the no-contact baseline during the contact window and converge to the same near-uniform final state afterward.

6.4 Parameter sweep

A sweep over (a, q_2) locates the historically motivated scenarios within the full parameter space and identifies the regions in which qualitatively different final-state behavior emerges. This is important because the canonical bifurcation analysis in Mitchener (2003) operates near $a = 0.5$, where coherent equilibria are accessible at high q and collapse as q falls. The historically motivated scenarios use higher values of a (0.75–0.80), reflecting a modeling assumption of relatively high cross-grammar intelligibility among related Germanic varieties. As the heatmap shows, this places the portfolio in a parameter region where the final concentration is near-uniform across the range of q_2 explored, meaning that the historically motivated contact scenarios do not approach a regime of bifurcation-driven final-state divergence. Lower values of a , closer to Mitchener’s canonical setting, produce a richer dependence of final M_2 on q_2 ; reducing a toward 0.5 while keeping q_2 in the moderate range is one direction in which future scenarios could explore a regime with more pronounced final-state consequences.

Table 4: Transient displacement integrals D_H (Eq. 10) and D_{M_2} (Eq. 11), computed by the trapezoid rule over $[0, 120]$ model units using `compute_transient_displacement()` in `language_dynamics_simulation_v3.py`.²The no-contact scenario has zero displacement from itself by definition and is omitted. Values of D_H for the conservative and moderate scenarios are small because normalized entropy in both the contact and no-contact conditions converges rapidly toward 1.0 under high- a regimes; D_{M_2} is less affected by this normalization effect and provides the clearer scaling signal among historically motivated scenarios.

Scenario	D_H	D_{M_2}
Canonical comparison	0.9912	3.5129
Conservative contact	0.0155	3.0058
Moderate contact	0.0170	3.0095
Strong contact	0.1747	3.0691

7 Discussion

The simulations illustrate how the symmetric language dynamical equation responds to contact-like perturbations. In the parameter regime used here—average intelligibility $a \in [0.75, 0.80]$, contact-phase fidelity $q_2 \in [0.55, 0.75]$ —all scenarios converge to the same near-uniform final state. This convergence is consistent with the bifurcation structure established in Mitchener (2003): in the parameter regimes explored here, the simulated trajectories approach the incoherent near-uniform equilibrium, and the contact phase produces a transient detour rather than a qualitative change in long-run behavior. The notable effect of contact is therefore transient rather than asymptotic: the reduction in q during the contact phase pushes the system toward a more distributed configuration, and the subsequent partial recovery of q allows the system to consolidate again, though along a trajectory that differs from the no-contact baseline during the contact window. This is consistent with the mechanism described in Mitchener (2003): a reduction in learning reliability increases the weight of the mutation-like term and can transiently displace the population from a coherent configuration, even when the long-run attractor is unchanged. The heatmap (Figure 7) shows that stronger disruption and eventual final-state differences are accessible at lower values of a and q_2 , but these lie outside the historically motivated parameter regime explored here.

However, the interpretation must remain cautious. The components $G_1, \dots, G_5$ are idealized grammatical systems, not directly observed populations. The parameters q , a , and the Old Norse fraction are not historical measurements. The model does not track specific grammatical features such as case loss, word order change, or agreement morphology. Therefore, it cannot establish that Old Norse contact caused Middle English simplification. It can only show that, within this mathematical framework, reductions in learning reliability have the right qualitative form to induce transient instability in the grammar distribution.

The main value of the model is thus exploratory. It offers a transparent way to formalize assumptions about learning reliability, contact, and grammatical competition, and to test whether qualitative outcomes are robust under parameter variation. Future work should connect the parameters to corpus-based proxies and should explore asymmetric intelligibility matrices.

8 Limitations

This study has several important limitations.

1. **Symmetric intelligibility.** The model assumes a single value a for all cross-grammar intelligibility relations. This is unrealistic for Old English dialects and Old Norse, but it

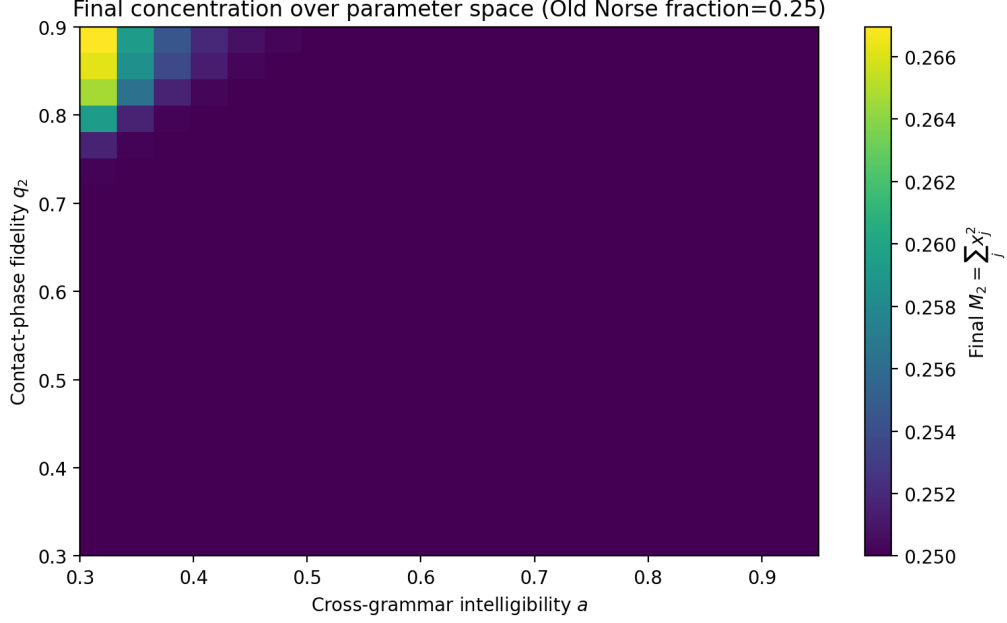


Figure 7: Heatmap of final M_2 under variation of average intelligibility a and contact-phase learning fidelity q_2 . The conservative, moderate, and strong contact scenarios use $a \in [0.75, 0.80]$ and $q_2 \in [0.55, 0.75]$, placing them in the near-uniform-final-state region of the upper-right portion of this space. The canonical comparison ($a = 0.5$) is a methodological reference at Mitchener’s canonical setting and lies further left in this space. Stronger final-state divergence emerges at lower a and lower q_2 .

preserves comparability with Mitchener’s analytical framework.

2. **Heuristic parameters.** The parameters are scenario values, not measurements. They must be supported by historical arguments and sensitivity analysis rather than treated as facts.
3. **No direct corpus calibration.** The present version does not estimate parameters from corpora. Corpus resources may provide indirect proxies in future work.
4. **No spatial structure.** The model assumes a freely interacting population, whereas historical language contact was geographically structured.
5. **No feature-level grammar.** The model tracks idealized grammar frequencies but does not represent particular morphosyntactic features.

9 Conclusion

This paper reinterprets the Old-to-Middle English transition as a computational case study for the symmetric language dynamical equation. Rather than proposing a new mathematical model or claiming to reconstruct historical demographics, it implements historically motivated scenarios suggested by Mitchener’s discussion of Scandinavian contact. The resulting simulations show how reductions in learning fidelity during a contact phase produce a measurable transient displacement in the diversity and concentration of model grammars, even when the long-run attractor is unchanged. The scalar displacement integrals D_H and D_{M_2} provide a compact quantitative summary of this transient effect across scenarios. The findings support the use of dynamical models as exploratory tools for historical linguistics, provided their assumptions are made explicit and their parameters are treated as scenario-based rather than directly observed.

Future work should strengthen the historical grounding with page-level primary-source references, corpus-informed proxies, and asymmetric intelligibility matrices.

Code Availability

All simulation code is publicly available at the project repository:

https://github.com/cmorregof/language_dynamics

The repository includes the main Python script, dependencies, and generated figures.

Declaration on AI-Assisted Tools

Claude (Anthropic) was used in an assistive capacity for:

- Language editing and grammar checking (no changes to scientific content or conclusions)
- Python code review for syntax and efficiency (no changes to algorithms or numerical methods)
- Literature organization and research planning

All mathematical derivations, numerical implementation, parameter choices, result interpretation, and conclusions are the sole work of the authors. All references were verified manually against primary sources.

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